Paramveer Multani

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Education

McMaster University Hamilton, Ontario
Bachelor of Computer Engineering (Co-op) *Expected Graduation Apr 2028*

- Relevant Coursework: Data Structures and Algorithms, Software Development, AI: Innovative Technologies, Advanced Probability and Random Processes, Digital Systems Design, Microprocessor Systems

Experience

Hoskin Scientific May 2026 – Sep 2026
Calibration & Test Engineering Intern *Oakville, ON*

- Cut calibration report prep from **10 minutes to under 1** across 20+ reports daily by building an offline **JavaScript** (ExcelJS) tool that parses instrument CSV exports into Excel worksheets.
- Trained a **machine learning** model in **Python** on historical calibration reports to validate incoming calibration data and flag out-of-spec readings before a report is signed off.
- Drove adoption at **all Canadian offices** by pitching both tools to management unprompted and writing the SOP technicians now follow.
- Restored 100+ scientific instruments to service through PCB-level diagnostics, resoldering or replacing failed parts.

BS Transportation Feb 2021 – Present
Operations & Systems Coordinator *Hamilton, ON*

- Recovered **50+ hours a year** by automating Avaal freight manifests with a Python script, cutting each from 5 min to under 2.
- Cut manual data entry by about 40% by building Excel tools for payroll and expense tracking, replacing a fully manual process.
- Reduced diagnostic time 25% and fleet downtime 20% by rewiring and reorganizing truck electrical harnesses to speed up fault isolation.

Projects

Emotion-Aware Companion Robot Oct 2026
Python, PyTorch, Transformers (RoBERTa, WavLM), Ollama, FastAPI, pytest

- Built a local robot that reads emotion from speech, voice, and face, then replies aloud **0.58 s** after the user stops talking, running fully on one PC with a 6 GB GPU.
- Fine-tuned **RoBERTa** on MELD to **0.638 weighted F1** on the held-out test set, about double the 0.313 always-neutral baseline, making every choice on dev and scoring test once.
- Fused text, WavLM audio, and vision heads with dev-tuned weights and calibrated confidence, so the robot checks in when signals conflict (a happy sentence in an angry voice) instead of guessing.
- Kept the full pipeline (VAD, ASR, emotion models, Ollama replies, TTS) at **2.18B parameters** under a 6B cap, covered by 206 CPU tests that run in about 15 s.

Hexcoach: Catan-Style Game Engine and AI Coach Sep 2026 – Present
Python, PyTorch, Monte Carlo Tree Search, Rust (PyO3), pytest, Hypothesis, GitHub Actions

- Trained a **PyTorch** value/policy network on self-play data generated in parallel with multiprocessing, used to guide a **Monte Carlo Tree Search** bot under a 2-second budget.
- Caught a seat bias (109/106/85 wins) that skewed every bot comparison and rebuilt the benchmarks with seat rotation and **95% confidence intervals**.
- Validated the engine with **147 tests** in CI and 6 invariants checked per move across 1,000 games (905K moves, 0 failures).
- Cut state copies to 0.98 μs per clone after profiling the search hot path, then ported it to Rust and verified it by replaying seeded games through both engines.

HousingHub: Student Housing Marketplace Feb 2025 – Present
TypeScript, React, Node.js, Express, PostgreSQL, Drizzle ORM, Socket.IO

- Modeled a normalized 12-table **PostgreSQL** schema in Drizzle ORM behind 24 REST endpoints, with Zod validators shared by client and server across 9,100+ lines of TypeScript.
- Replaced raw distance with walk, bike, drive, and transit commute times per listing by integrating the OpenRouteService and TravelTime APIs.
- Delivered real-time landlord and student messaging over Socket.IO with conversation rooms, read receipts, and unread counts.

Indoor LIDAR Spatial Mapping System Jan 2026 – Apr 2026
Python, Bare-Metal C, ARM Cortex-M4F, VL53L1X, I2C, UART

- Built a Python pipeline that turns 64 time-of-flight readings per sweep into 3D point clouds, converting polar measurements streamed over I2C and UART from bare-metal C firmware.
- Found the throughput bottleneck by profiling each stage: the stepper took **640 ms per point** against 32 ms of bus overhead.

Technical Skills

Languages: Python, C, C++, TypeScript, JavaScript, SQL, Rust
AI/ML: PyTorch, Transformers (RoBERTa, WavLM), Monte Carlo Tree Search, Ollama, Machine Learning
Backend & Web: FastAPI, Node.js, Express, REST APIs, PostgreSQL, React, React Native, Socket.IO, Firebase, Supabase
Embedded & Hardware: ARM Cortex-M4F, Bare-Metal C, I2C, UART, PCB Diagnostics, Soldering
Testing & Tools: pytest, Hypothesis, GitHub Actions CI, Git, Linux
Interests: Basketball, Chess, Hiking, Sports Cars